NLO Gluon Production: DGLAP, Running-Coupling, JIMWLK and BFKL Pieces

An exact rewriting of the NLO single-gluon spectrum and its separation into evolution terms

$d_\perp = 2-\epsilon$ (so $2/\epsilon = 1/\hat{\epsilon}$ for $D = 4-2\hat{\epsilon}$), $\overline{\rm MS}$ scale $\mu \equiv \mu_{\overline{MS}}$, pure glue ($n_f = 0$).

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0. Starting expression

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\Big]\\
&\quad\times\Big[\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big)\Big]\,\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\cdots\Big]\,\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)}d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)/\xi}\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O}
\end{aligned}$$

where $[\cdots]$ in the second line is the same bracket as in the first, and $\mathcal{O} = [U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ec}(x)]\,\rho^{e'}(x')\rho^{e}(x)$.

> In the original expression the last Wilson line reads $U^{ce}(x)$. It is taken here to be $U^{ec}(x)$; otherwise that factor is $U^\dagger(x)$.

1. Notation

$$r \equiv y'-y, \qquad \mathcal{N} \equiv \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{1}{k^+}, \qquad \frac{g^2N_c}{2\pi^2} = 2\bar{\alpha}_s, \quad \bar{\alpha}_s = \frac{\alpha_s N_c}{\pi}, \qquad a \equiv \frac{k^+}{\vee}, \quad b \equiv \frac{k^+}{k^+\!+\Lambda},$$

$$\mathcal{A} = \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}, \qquad \mathcal{B} = \frac{(x-y)\cdot r\,(x'-y')\cdot r}{(x-y)^2(x'-y')^2\,r^2}, \qquad \Phi_W(q) \equiv \int_{x,y,x',y'} e^{-iq\cdot r}\,W\,\mathcal{O}.$$

$\Phi_{\mathcal{A}}(q)$ is the LO structure (amplitude times conjugate) evaluated at transverse momentum q.

2. Two exact preliminaries

2.1 The bracket L

Pull out $d_\perp/2$:

$$L \equiv -\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{r^2\mu^2}{4} - 1 = -\frac{d_\perp}{2}\Big[\frac{2}{\epsilon}+\log\frac{r^2\mu^2}{c_0^2}\Big] - 1, \qquad c_0 \equiv 2e^{-\gamma_E}.$$

Using $d_\perp = 2-\epsilon$, the $O(\epsilon)$ part of $d_\perp/\epsilon$ gives $+1$, which cancels the -1 exactly:

$$\boxed{L = -\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2} + O(\epsilon).}$$

The $O(\epsilon)$ terms can be dropped because L only multiplies finite quantities: the ξ-integrals at fixed $\vee, \Lambda$, and a $g \to gg$ splitting function that is ϵ-independent. Equivalently,

$$L = 4\pi\int\frac{d^{d_\perp}q}{(2\pi)^{d_\perp}}\frac{e^{iq\cdot r}}{q^2} \;\simeq\; \frac{1}{\pi}\int d^{d_\perp}z\,\frac{(z-y)\cdot(z-y')}{(z-y)^2(z-y')^2}.$$

So the pole is a final-state collinear one: relative transverse momentum $q \to 0$, i.e. a large separation of the unobserved gluon at z, with no Wilson line $U(z)$.

2.2 The blue bracket

$$\log\frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+}-1\Big) = \log\frac{\vee-k^+}{\Lambda} = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} + \log\Big(1-\frac{k^+}{\vee}\Big).$$

The last piece is $O(k^+/\vee)$; it is not a rapidity log.

3. Exact rewriting

All three lines share the same prefactor $2\bar{\alpha}_s\mathcal{N}$. In the third line $d_\perp \to 2$, because it has no pole.

$$\boxed{\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= 2\bar{\alpha}_s\mathcal{N}\Big\{\underbrace{\int_a^1\frac{d\xi}{\xi^2}\,p(\xi)\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big)}_{\text{(I)}} + \underbrace{\log\frac{\vee-k^+}{\Lambda}\,\Phi_{\mathcal{A}L}(k)}_{\text{(II)}} + \underbrace{\int_a^b d\xi\,\frac{1-\xi}{\xi}\,\Phi_{\mathcal{A}-2\mathcal{B}}\big(\tfrac{k}{\xi}\big)}_{\text{(III)}}\Big\}\\
p(\xi) &= \xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}.
\end{aligned}}$$

The full pure-glue splitting function, with $\beta_0 = \frac{11}{3}N_c$, is

$$P_{gg}(\xi) = 2N_c\, p(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi) \qquad \Longrightarrow \qquad p(\xi) = \frac{P_{gg}(\xi)}{2N_c} - \frac{11}{12}\,\delta(1-\xi).$$

The momentum sum rule confirms the coefficient:

$$\int_0^1 d\xi\,\xi\,p(\xi) = -\tfrac{3}{2} + \tfrac{1}{2} + \tfrac{1}{12} = -\tfrac{11}{12}, \qquad \text{so} \qquad \int_0^1 d\xi\,\xi\,P_{gg} = 0.$$

Adding and subtracting the $\delta(1-\xi)$ term in (I) gives the separation below.

4. DGLAP: line (I), completed to the full P_{gg}

$$\frac{d^3N^{\mathrm{DGLAP}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\,P_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot r/\xi}\,\mathcal{A}\Big[-\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

- Here $(1-\xi)_+$ is the standard plus distribution on $[0,1]$.
- If the plus distribution in the original expression is defined on $[k^+/\vee, 1]$, the $\log(1-k^+/\vee)$ piece of the blue bracket converts it to the $[0,1]$ one exactly: $\int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-a)$, with $f(\xi) = e^{-ik\cdot r/\xi}/\xi$ and $f(1) = e^{-ik\cdot r}$.
- If it was already the $[0,1]$ one, that piece instead stays as an $O(k^+/\vee)$ leftover (Section 8).
- This is the timelike $g \to g$ fragmentation convolution: $\frac{d\xi}{\xi^3}$ times $\frac{dN_{\mathrm{LO}}}{d^3p} \propto \frac{1}{p^+}$ at $p^+ = k^+/\xi, p = k/\xi$.
- The $-2/\epsilon$ is removed by the $\overline{\mathrm{MS}}$ counterterm of $D_g(\xi,\mu^2)$. The natural scale is $\mu = c_0/r$, i.e. $\mu \sim k$ in momentum space.

5. Running coupling: the subtracted $\delta(1-\xi)$ piece

$$\frac{d^3N^{\mathrm{RC}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{1}{k^+}\frac{11N_c}{6}\int_{x,y,x',y'} e^{-ik\cdot r}\,\mathcal{A}\Big[\frac{2}{\epsilon} + \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O} = \frac{\alpha_s\beta_0}{2\pi}\mathcal{N}\Big[\frac{2}{\epsilon}\,\Phi_{\mathcal{A}}(k) + \Phi_{\mathcal{A}\log(r^2\mu^2/c_0^2)}(k)\Big]$$

- In pure dimensional regularization the on-shell self-energy is scaleless, so its $1/\epsilon_{\mathrm{UV}}$ and $1/\epsilon_{\mathrm{IR}}$ cancel. That lets this pole be read as the UV pole removed by coupling renormalization of the LO.
- The log then makes the LO coupling run to the dipole scale:

$$\alpha_s(\mu^2) \;\to\; \alpha_s\big(c_0^2/r^2\big) = \alpha_s(\mu^2)\Big[1 + \frac{\alpha_s\beta_0}{4\pi}\log\frac{r^2\mu^2}{c_0^2}\Big].$$

- With quarks, $\beta_0 \to \frac{11N_c - 2n_f}{3}$, and the $g \to q\bar{q}$ real channel must also be present.

6. JIMWLK: the term proportional to $\log(k^+/\Lambda)$

$$\frac{d^3N^{\mathrm{JIMWLK}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2N_c}{2\pi^2}\frac{1}{k^+}\,\log\frac{k^+}{\Lambda}\int_{x,y,x',y'} e^{-ik\cdot r}\,\mathcal{A}\Big[-\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

This comes from the soft end $\xi \to 1$: the unobserved gluon has $p^+ - k^+ = k^+(1-\xi)/\xi$, which reaches the cutoff Λ at $\xi = b$.

7. BFKL: the term proportional to $\log(\vee/k^+)$

$$\frac{d^3N^{\mathrm{BFKL}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2N_c}{2\pi^2}\frac{1}{k^+}\,\log\frac{\vee}{k^+}\int_{x,y,x',y'} e^{-ik\cdot r}\,\mathcal{A}\Big[-\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

This comes from the end $\xi \to k^+/\vee$, where the parent momentum $p^+ = k^+/\xi$ reaches $\vee$. See also Section 9, item 3: line (I) contains an additional $\log(\vee/k^+)$ term.

8. What is left over

8.1 The $\log(1-k^+/\vee)$ term

$$2\bar{\alpha}_s\mathcal{N}\,\log\Big(1 - \frac{k^+}{\vee}\Big)\,\Phi_{\mathcal{A}L}(k).$$

Depending on the plus-prescription convention, this either becomes the endpoint term in Section 4 or stays as a power correction.

8.2 Line (III) is finite and has no rapidity log

- **Upper end.** The factor $(1-\xi)$ vanishes at $\xi \to 1$, so setting $b \to 1$ costs only $\int_b^1 (1-\xi)\,d\xi = O(\Lambda^2/k^{+2})$. No $\log(k^+/\Lambda)$.

- **Lower end.** As $r \to 0$, the r-dependence of $\mathcal{A} - 2\mathcal{B}$ is the traceless combination $(\delta_{ij} - 2\hat{r}_i\hat{r}_j)$, whose angular average vanishes. Its Fourier transform is

$$\int d^2r\, e^{-iq\cdot r}\,(\delta_{ij} - 2\hat{r}_i\hat{r}_j) = -\frac{4\pi}{q^2}\,(\delta_{ij} - 2\hat{q}_i\hat{q}_j),$$

so $\Phi_{\mathcal{A}-2\mathcal{B}}(k/\xi) \sim \xi^2/k^2$. The integrand $\frac{1-\xi}{\xi}\,\Phi$ then goes like ξ, and setting $a \to 0$ costs only $O(k^{+2}/\vee^2)$. No $\log(\vee/k^+)$.

$$\frac{d^3N^{\text{finite}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2N_c}{2\pi^2}\frac{1}{k^+}\int_0^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot r/\xi}\big[\mathcal{A} - 2\mathcal{B}\big]\mathcal{O} + O\Big(\frac{\Lambda^2}{k^{+2}}, \frac{k^{+2}}{\vee^2}\Big)$$

This is the genuine NLO impact-factor term.

8.3 Bookkeeping

Original piece	Equals
Line (I)	DGLAP + RC
Line (II)	JIMWLK + BFKL + $\log(1 - k^+/\vee)$ term
Line (III)	finite remainder

The separation is exact up to $O(\epsilon)$.

9. Things to check

1. **A factor of 2 in normalization.** Relative to $\mathcal{N}\Phi_{\mathcal{A}}$, the DGLAP coefficient above is $(\alpha_s/\pi)P_{gg}$ and the RC coefficient is $\alpha_s\beta_0/2\pi$. Both are the textbook values, $(\alpha_s/2\pi)P_{gg}$ and $\alpha_s\beta_0/4\pi$, only if the LO is

$$\frac{dN_{\text{LO}}}{d^3k} = 2\,\mathcal{N}\,\Phi_{\mathcal{A}}(k).$$

   The two checks agree with each other. If the LO is $\mathcal{N}\Phi_{\mathcal{A}}$, the NLO prefactor has a stray factor of 2, for example from the identical-gluon factor $\frac{1}{2}$.
2. **The JIMWLK and BFKL pieces carry L, including its $1/\epsilon$.** They are soft–collinear double-log terms: a rapidity log times the final-state collinear pole. In the full NLO, $\frac{1}{\epsilon}\log\frac{k^+}{\Lambda}$ should cancel against virtual corrections with the same cutoff Λ; the $U(z)$ terms of the evolution kernel are not in the expression shown. Do not also absorb these poles into DGLAP, or they will be double counted. The finite evolution kernels are the $\log(r^2\mu^2/c_0^2)$ parts.
3. **Line (I) hides some $\log(\vee/k^+)$ dependence.** The $(1-\xi)/\xi^3$ term near $\xi \to k^+/\vee$ picks up the large-q tail

$$\Phi_{\mathcal{A}L}(q) \xrightarrow{q\to\infty} \frac{4\pi}{q^2}\,\Sigma, \qquad \Sigma \equiv \int_{x,x',y} \mathcal{A}\,\mathcal{O}\Big|_{y'=y},$$

   using $\int d^2r\, e^{-iq\cdot r}(-\log r^2) = 4\pi/q^2$ for $q \neq 0$ and $\int_{y'} = \int d^2y'$. (The pole part and the $\log\mu^2$ part fall off faster.) This contributes

$$2\bar{\alpha}_s\,\mathcal{N}\,\frac{4\pi}{k^2}\,\Sigma\,\log\frac{\vee}{k^+},$$

   which is exactly the large-k limit of the BFKL term in Section 7. So the lower limit in line (I) cannot be set to 0, and if all of the $\log(\vee/k^+)$ dependence is wanted, this term has to be moved into the BFKL piece.